



Transfer data to a new phone

Buying a new phone but don't know how to transfer all your data over? Here's a detailed tutorial. <https://digit.in/dec19-32>



Google Stadia on rooted Android

The folks over at XDA-Developers have found a way for rooted Android phones to access Google Stadia. <https://digit.in/dec19-33>

Intel Core i9-10980XE Extreme Edition

More of the same, unless you're into AI

Intel is revising their HEDT line-up with a brand new 14nm silicon in the form of the Intel Cascade-X Core i9-10980XE. We're obviously using the term "brand new" with a hint of sarcasm. Based on the same X299 platform which was unveiled at COMPUTEX 2017, Intel has released three line-ups which began with Skylake-X 7980XE, followed by the Skylake-X Refresh 9980XE and now we have the Cascade Lake-X 10980XE. While the cores are very much similar to Skylake-X, there are a few changes which make Cascade Lake-X stand out. However, the good part is that because the architecture is very much the same, you can still use the existing X299 platform, making this an easy upgrade. The highlight is that all of this is now available at roughly half the cost of what Intel's previous HEDT flagship used to cost. Let's see how it fares.

What's new

As per the spec sheet, the Turbo Boost 3.0 frequency is higher by 300 MHz at 4.8 GHz. Then there's support for 256 GB of DDR4-2933 MHz RAM. There are four extra PCIe lanes, and support for DL Boost. We have to appreciate the improvements in clock speeds that Intel has managed to achieve over the years with the same core, however, if we are to look at the 3970X, then they've managed to hit 3.7 GHz base clock for 32 cores as opposed to 3.0 GHz for 18 cores that Intel has managed to. During peak loads, both processors draw ridiculous amounts of power. The DL-Boost addition is a welcome feature as we've seen a lot of software incorporate AI elements.

Rendering Performance

In Blender, POV-Ray and Corona 1.3, we could hardly see any difference in performance. Processors with higher core



PERFORMANCE.....84
VALUE FOR MONEY....75

counts outperform it. Then, there are software such as Handbrake where Intel still maintains a lead over AMD. In our Blender benchmark, the Threadripper 2990WX finished the run in the

fastest time, followed by the 2970WX and then the Intel Core i9-9980XE and 10980XE. The difference between the no.1 spot and the 10980XE is approximately 42 seconds, similar to what we saw with the 9980XE. There's still AVX512, which gives Intel an advantage, however, there's a multiplier offset that kicks in whenever AVX512 is utilised. For prolonged workloads that make use of AVX512, there is a performance hit experienced. And as for DL-Boost, you can take advantage of it if you're using TensorFlow, OpenVINO or PyTorch to build AI models.

Gaming Performance

Coming to gaming, the HEDT processors are still behind their mainstream desktop counterparts. The 9900K/KS is still the best gaming processor but more and more games are coming out where the performance gap is narrow. We wouldn't chalk this up to HEDT processors getting better, rather the fact that game developers are switching to engines that make better use of multithreaded processors. Shadow of War is one such game where the

processor hardly matters. It's when you scale down to a lower resolution that you see a difference in performance. In comparison to the 9980XE, we don't see a noteworthy change.

Power consumption

Our power consumption values haven't changed compared to the 9980XE. It's still consuming about 195 Watts at peak load which is, coincidentally, close to the mainstream 9900KS. If we were to put out a watt/core metric then the 10980XE is averaging about 10.83 watts/core. So if your primary task scales evenly (it never does) with respect to core count across AMD and Intel CPUs, then AMD is clearly the economical choice over the long run. However, should the same software be capable of using DL-Boost to accelerate the workload, then the tables are easily turned. Since the overall power situation has remained mostly the same, you can retain your existing X299 motherboard.

Bottomline

With Cascade Lake-X, Intel's pricing strategy has changed for the better. We're seeing the same core count at half the price with the added perks of better memory support and improved clock speeds. The added AI acceleration is a plus for developers working with AI models. This would make AMD rethink their pricing model to compete at a lower price point. However, AMD has the core count advantage which it might still choose to sell at a premium. Overall, the Intel Core i9-10980XE is certainly a much better value proposition in the current HEDT segment.

—Mithun Mohandas

SPECIFICATIONS

CLOCK SPEED: 3.0 GHz | TURBO FREQUENCY: 4.8 GHz | CORES/THREADS: 18/36 | LITHOGRAPHY: 14nm | TDP: 165W | PCIe LANES: 48 | PCIe VERSION: 3.0 | MEMORY SUPPORT: DDR4-2933

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